

Chocolate Sales Analysis Report — Part 6 (Power Pivot & Dashboards)

1. Introduction

This report documents my full workflow for completing **Chocolate Sales** project, which required building a **Power Pivot Data Model**, creating **DAX measures**, producing **multiple PivotTable analysis**, and designing **interactive dashboard sections**.

The analysis focuses on sales performance using four perspectives:

1. **Overall Sales Trend & KPIs**
2. **Product Analysis**
3. **Salesperson Analysis**
4. **Time & Region Analysis**

Screenshots of all dashboards and pivot tables are inserted throughout the report.

2. Data Model Construction

✓ Data Sources Loaded

I imported the following tables into the Data Model using **Power Query**:

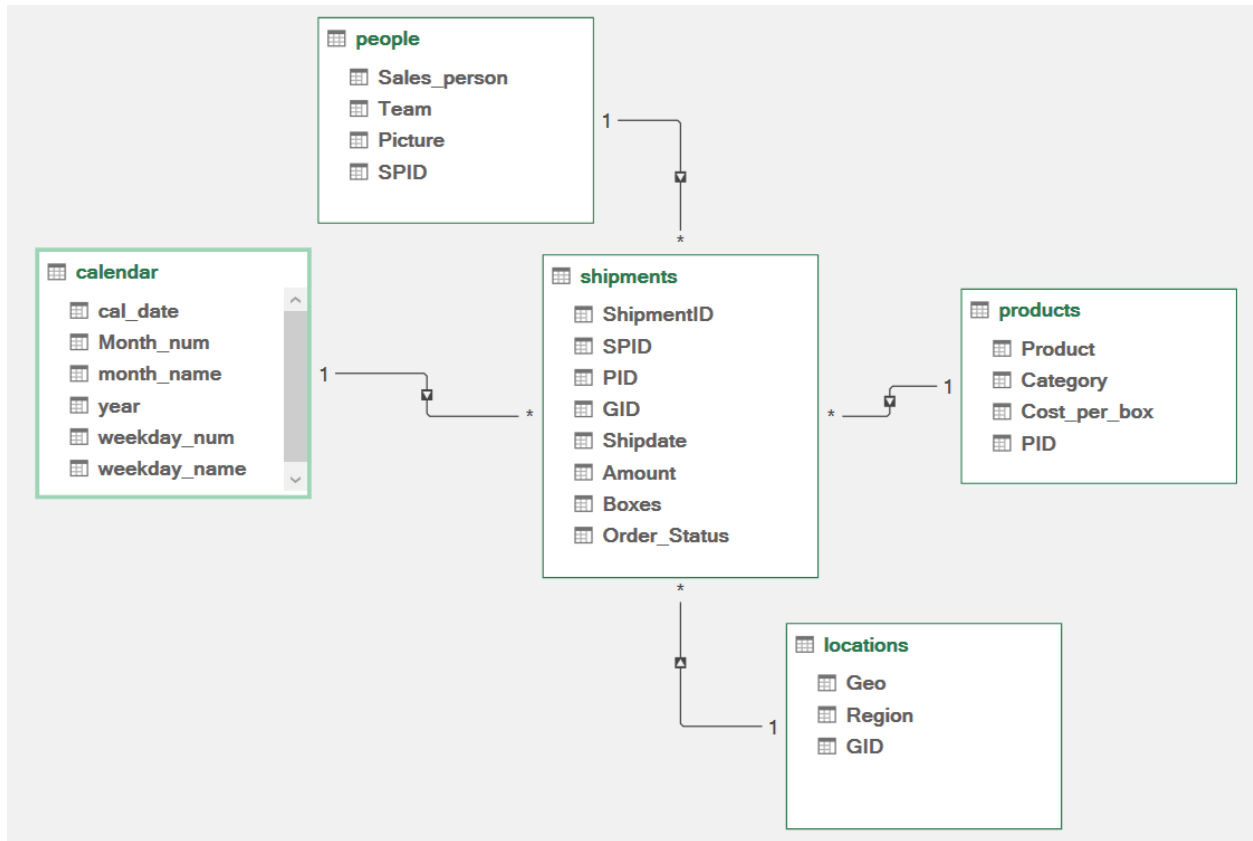
- **shipments**
- **products**
- **people**
- **locations**
- **calendar**

✓ Relationships Created (Star Schema)

Fact table → **shipments**

Dimension tables → products, people, locations, calendar

Final Relationship Diagram



Explanation:

- **calendar[cal_date] 1 → M shipments[Shipdate]**
- **products[PID] 1 → M shipments[PID]**
- **people[SPID] 1 → M shipments[SPID]**
- **locations[GID] 1 → M shipments[GID]**

This structure supports slicing by **time, salesperson, product, and region**.

3. DAX Measures Created

File Home Design Advanced									
Clipboard		Get External Data		Formatting		Sort and Filter		Find	
Paste	Paste Append	From Database	From Service	From Other Sources	Existing Connections	Refresh	PivotTable	Data Type	Format
Paste	Paste Replace							Sort A to Z	Sort Z to A
Copy								Clear Sort	Clear All Filters
								Sort by Column	Find
								AutoSum	Create KPI
								Data View	Diagram View
								Show Hidden	Calculation Area

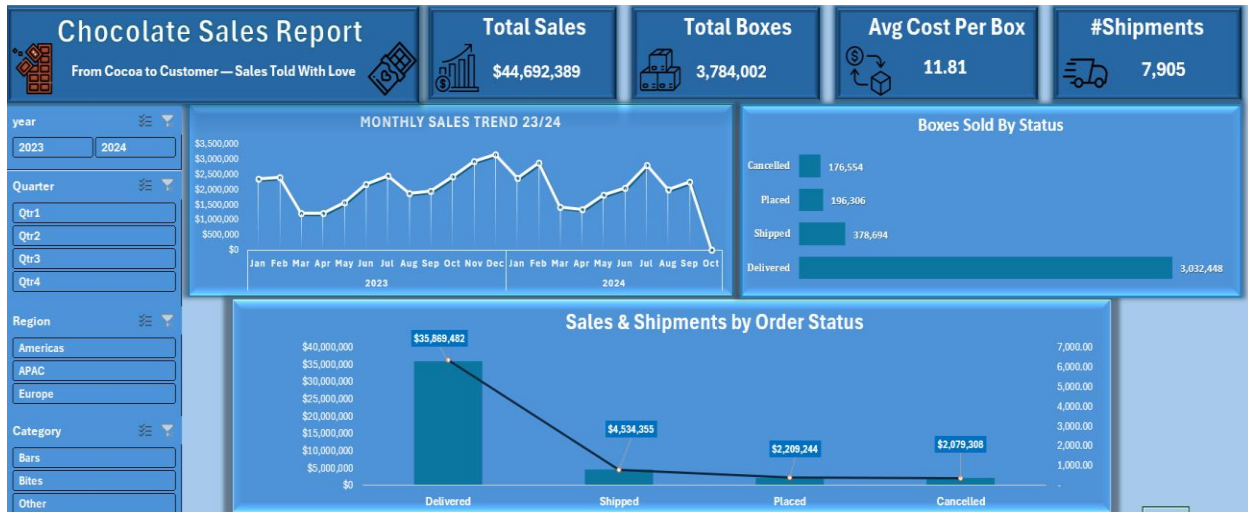
ShipmentID	S...	P...	G...	Shi...	Amount	Boxes	Order Status
1	S00000593	SP01	P10	G6	10/11/202...	1372.5	99 Delivered
2	S00002124	SP01	P10	G6	5/17/2023 ...	920.25	66 Delivered
3	S00003637	SP01	P10	G6	2/5/2024 1...	4432.5	317 Delivered
4	S00003722	SP01	P10	G6	2/22/2023 ...	823.5	64 Delivered
5	S00005183	SP01	P10	G6	4/17/2024 ...	2574	198 Delivered
6	S00008276	SP01	P10	G6	6/16/2023 ...	452.25	33 Delivered
7	S00008292	SP01	P10	G6	7/25/2023 ...	7341.75	490 Delivered
8	S00009808	SP01	P10	G6	8/29/2023 ...	1017	64 Delivered
9	S00009823	SP01	P10	G6	2/13/2023 ...	2711.25	181 Delivered
10	S00011354	SP01	P10	G6	1/23/2023 ...	18117	1295 Delivered
11	S00012849	SP01	P10	G6	3/14/2023 ...	1293.75	87 Delivered
12	S00014346	SP01	P10	G6	3/7/2023 1...	3411	244 Delivered
13	S00017458	SP02	P10	G6	11/28/202...	3645	243 Delivered
14	S00018988	SP02	P10	G6	5/13/2024 ...	429.75	31 Delivered
15	S00019019	SP02	P10	G6	4/5/2023 1...	562.5	47 Delivered
16	S00020532	SP02	P10	G6	2/8/2023 1...	4497.75	346 Delivered
17	S00023525	SP02	P10	G6	12/25/202...	4671	334 Delivered
18	S00028134	SP03	P10	G6	6/16/2023 ...	1885.5	118 Delivered
19	S00029715	SP03	P10	G6	1/5/2024 1...	6412.5	401 Delivered
20	S00029746	SP03	P10	G6	2/15/2024 ...	7364.25	567 Delivered
21	S00031277	SP03	P10	G6	5/28/2024 ...	3021.75	189 Delivered
					Total Sales Amount: \$44,692.389	Total Boxes: 3,784.002	
					Number of Shipments: 7905	Average Cost per Box: 11.81	

DAX List & Meaning

Measure	Expression	Purpose
Total Sales Amount	=SUM(shipments[Amount])	Total revenue
Total Boxes	=SUM(shipments[Boxes])	Total boxes sold
Number of Shipments	=COUNT(shipments[ShipmentID])	Total shipment count
Average Cost per Box	=[Total Sales Amount] / [Total Boxes]	Revenue efficiency

These measures were used across all dashboards to maintain consistent, accurate KPI calculations.

4. Section 1 — Overall Sales Trend & KPIs



PivotTables Created

1. **Monthly Sales Trend**
 - Shows seasonality and peaks.
2. **Boxes Sold by Status**
3. **Sales & Shipments by Order Status**

Charts Used

- Line chart → Monthly Sales Trend
- Bar chart → Boxes Sold by Status
- Combo chart → Sales vs Shipments by Status

KPIs Displayed

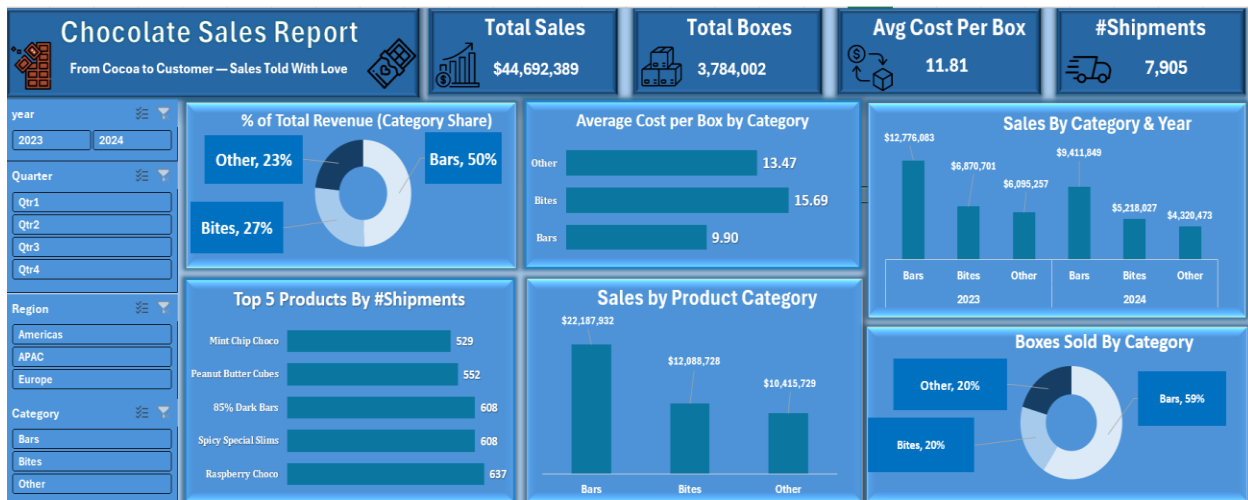
- Total Sales
- Total Boxes
- Avg Cost / Box
- #Shipments

Key Insights

- **Delivered** orders dominate sales: **\$35.8M** revenue.

- **Q1** is the strongest quarter with over **\$12M** in sales.
- Sales peak in **June and August**, while **April and October** show drops.
- Over **3M boxes** were delivered — the company’s main operational workload.

5. Section 2 — Product Analysis



PivotTables Created

1. Sales by Product Category
2. Average Cost per Box by Category
3. Top 5 Products by #Shipments
4. Sales by Category & Year
5. Boxes Sold by Category

Charts Used

- Pie chart → Revenue Share by Category
- Bar chart → Sales by Category
- Bar chart → Avg Cost per Box
- Bar chart → Top Products by Shipments
- Pie chart → Boxes Sold by Category

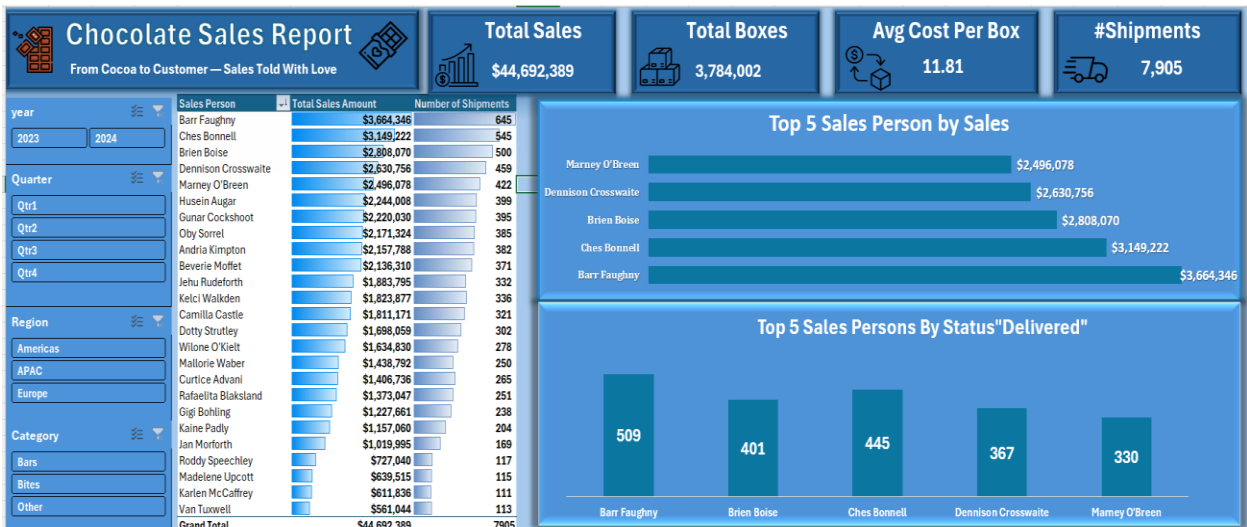
KPIs

- Revenue share by category
- Avg cost per category
- Best-performing products

Key Insights

- **Bars** represent **50%** of total revenue → clearly the flagship category.
- **Bites** category shows the **highest cost per box**, indicating premium pricing.
- Top-selling products include **Mint Chip Choco** and **Peanut Butter Cubes** with over **500 shipments each**.
- Sales are stable across 2023 and 2024 with Bars consistently leading.

6. Section 3 — Salesperson Performance



PivotTables Created

1. Top 5 Salespersons by Total Sales
2. Top 5 Salespersons by Delivered Status Shipments
3. Full Salesperson Ranking Table

Charts Used

- Horizontal bar chart → Revenue by Salesperson
- Column chart → Delivered Shipments Ranking

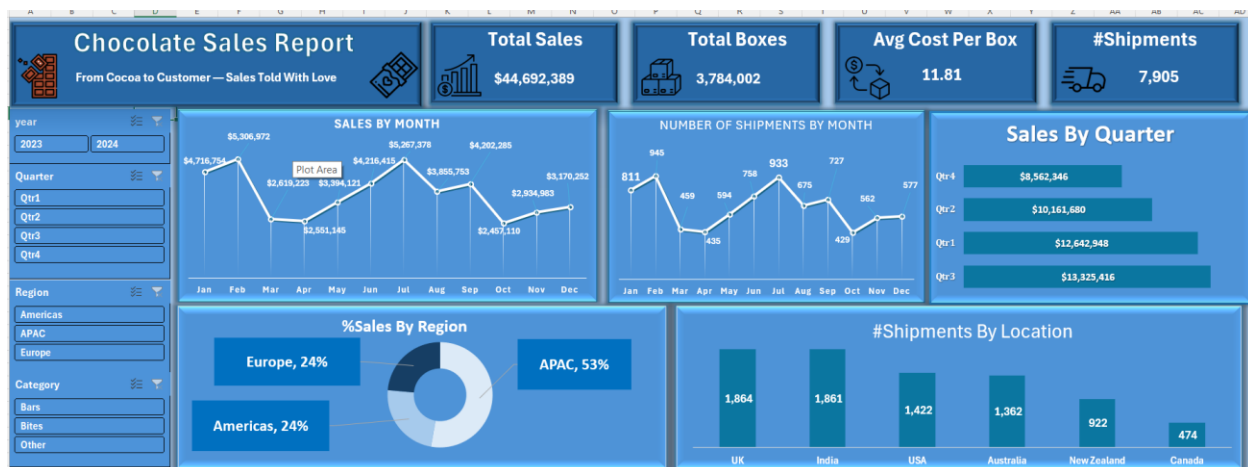
KPIs

- Highest revenue salesperson
- Most reliable deliverer
- Number of shipments per person

Key Insights

- **Barr Faughny** leads with **\$3.6M** in sales and **645** shipments.
- The top 5 account for a large share, showing a **cluster of high performers**.
- Delivered-status ranking matches revenue leaders, demonstrating consistency.

7. Section 4 — Time & Region Analysis



PivotTables Created

1. Sales by Month
2. Shipments by Month
3. Sales by Quarter
4. Shipments by Location

5. Sales by Region

Charts Used

- Line chart → Monthly shipments
- Bar chart → Quarterly revenue
- Column chart → #Shipments by Location
- Pie chart → %Sales by Region

KPIs

- Region distribution
 - Monthly shipment capacity
 - Quarterly sales trends
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Key Insights

- **APAC** leads with **53% of total sales** → strongest region.
 - **Europe** and **Americas** follow evenly, each at **24%**.
 - **India, UK, and USA** are top shipment destinations.
 - Shipments drop in April and October → operational bottlenecks.
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8. Reflection Paragraph

Working on Part 6 significantly strengthened my understanding of **Power Pivot, Data Modeling, and DAX**. This was my first time designing a full **star-schema model** and building interactive dashboards driven by DAX calculations. I learned how important data relationships are for accurate slicing and filtering, and I gained confidence in using DAX measures to produce consistent KPIs across multiple PivotTables. Designing four dashboards pushed me to think like a BI analyst — identifying key metrics, choosing the right chart types, and extracting meaningful insights. Overall, the project improved my ability to turn raw data into a professional, interactive report with real business value.

9. Final Insights Summary:

- **Revenue Strength:** \$44.6M total sales across all dimensions shows strong business stability.
- **Category Leadership:** Bars are the top-selling and highest-impact category.
- **Region Strategy:** APAC is the most profitable region — marketing should prioritize it.
- **Salesperson Impact:** Five top performers drive a huge share of company revenue.
- **Operational Insight:** Delivered orders dominate; cancelled orders still significant (176k boxes).
- **Seasonality:** Clear monthly peaks show predictable demand cycles for inventory planning.

10. Business Recommendations

Section 1 — Overall Sales Trend & KPIs

Findings:

- Delivered orders generate \$35.8M
- Q1 strongest quarter
- Seasonal peaks in June & August
- Drops in April & October

Business Recommendations:

1. **Increase production and inventory before seasonal peaks**
 - Prepare stock for June–August to avoid shortages during high demand.
2. **Launch targeted promotions during low seasons**
 - Discount campaigns or bundle offers in April and October.
3. **Optimize delivery operations**
 - Delivered orders drive most revenue → strengthen logistics partnerships.
4. **Re-examine operational planning for Q2–Q4**
 - Variations suggest room for forecasting improvements.

Section 2 — Product Analysis

Findings:

- Bars = 50% of total revenue
- Bites have highest cost per box (premium pricing)
- Best-selling products: Mint Chip Choco, Peanut Butter Cubes
- Consistent category performance

Business Recommendations:

1. **Prioritize the Bars product line**
 - Increase production, marketing spend, and distribution reach.
 2. **Introduce premium packaging or bundles for Bites**
 - Higher cost per box allows margin expansion.
 3. **Expand successful SKUs like Mint Chip Choco**
 - Consider new flavors based on customer interest.
 4. **Evaluate underperforming products**
 - Rebrand, reformulate, or discontinue if margins are low.
 5. **Plan cross-year inventory**
 - Category performance is stable → long-term forecasting becomes easier.
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Section 3 — Salesperson Performance

Findings:

- Barr Faughny leads with \$3.6M and 645 shipments
- Top 5 reps generate most revenue
- Delivered vs. revenue performance highly correlated

Business Recommendations:

1. **Reward and retain top-performing salespeople**
 - Bonuses, commissions, or leadership opportunities.

2. Analyze winning behaviors of top 5 reps

- Use insights for training underperforming employees.

3. Optimize salesperson territory allocations

- High performers may expand to additional regions.

4. Performance-based coaching

- Identify reps with low delivered shipments → targeted coaching.

Section 4 — Time & Region Analysis

Findings:

- APAC accounts for 53% of revenue
- Europe & Americas = 24% each
- Top shipment countries: India, UK, USA
- Shipment dips in April & October

Business Recommendations:

1. Expand market presence in APAC

- More distributors, local marketing, and region-focused campaigns.

2. Develop growth strategy for Europe and Americas

- Large potential for scaling as performance is balanced but not dominant.

3. Deepen partnerships in India, UK, USA

- These markets show strong shipping activity → potential for loyalty programs.

4. Investigate low-shipment months

- Understand if dips are caused by supply chain issues or demand fluctuations.

5. Diversify regional risk

- Heavy reliance on APAC could be risky → grow emerging regions.